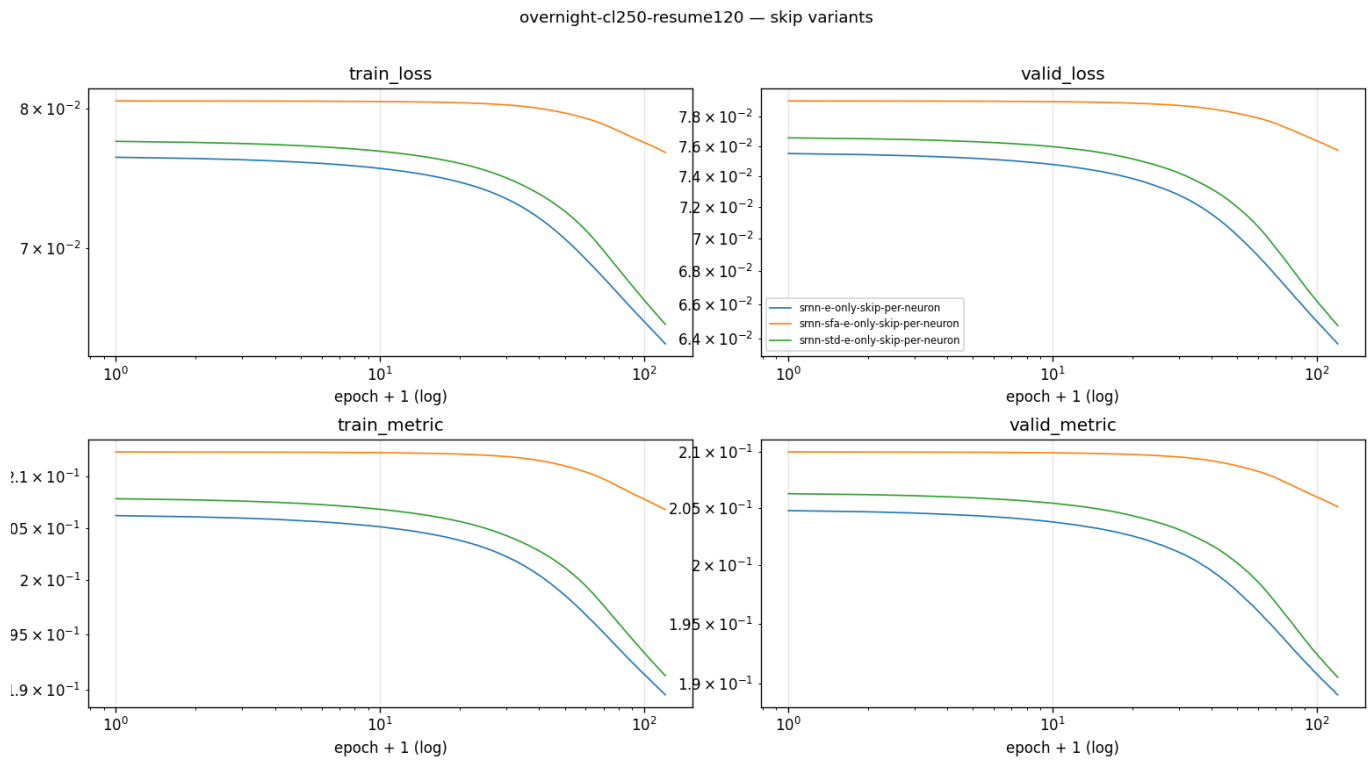


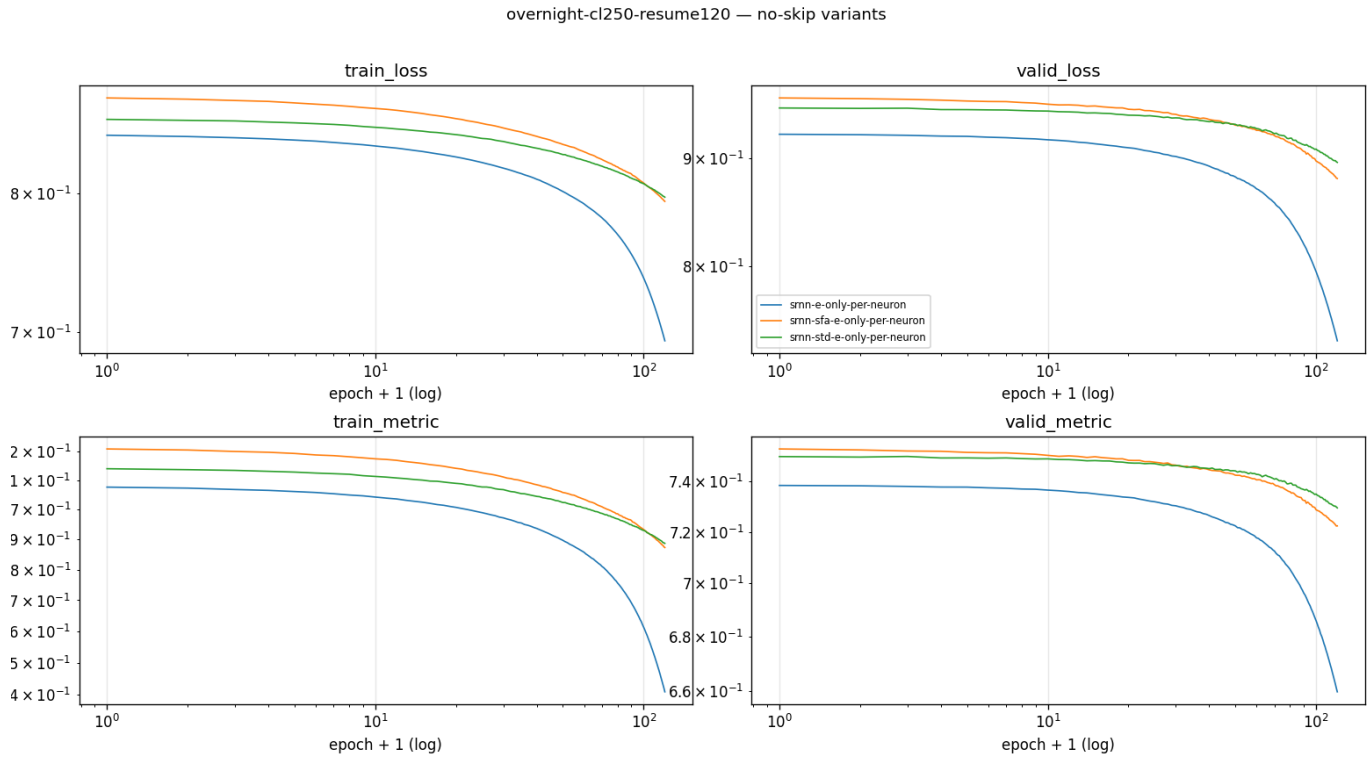
Run: overnight-cl250-resume120

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07435 → 0.06201
 srnn-sfa-e-only-skip-per-neuron: 0.07804 → 0.07453
 srnn-std-e-only-skip-per-neuron: 0.07543 → 0.06321

Figure 1: skip variants — log-log loss/metric



Final test loss (init - last) [no-skip]:
 srnn-e-only-per-neuron: 0.82829 - 0.66636
 srnn-sfa-e-only-per-neuron: 0.85538 - 0.79321
 srnn-std-e-only-per-neuron: 0.84749 - 0.80718

Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-e-only-per-neuron)

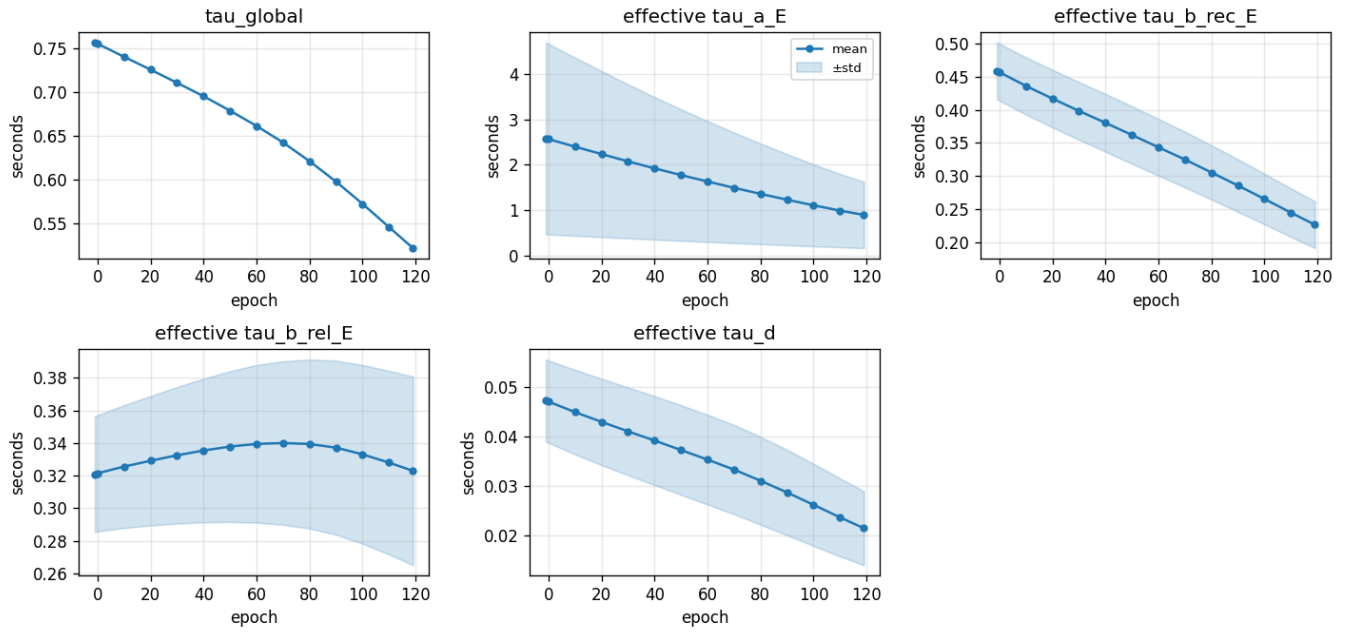
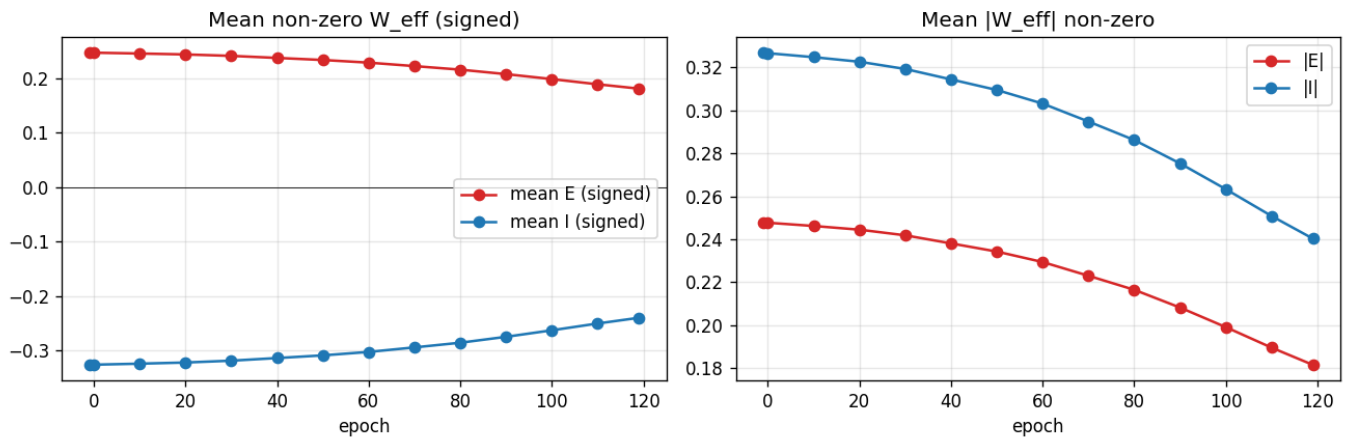


Figure 3: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# overnight-cl250-resume120 - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+0.75694	+0.52182	-	-	scalar
tau_d (s)	+0.047285	+0.021479	0.00829	0.007465	300
tau_a_E (s)	+2.5842	+0.89414	2.116	0.7305	450
tau_b_rec_E (s)	+0.45844	+0.22725	0.04265	0.0352	150
tau_b_rel_E (s)	+0.32092	+0.32289	0.03539	0.05786	150
W_eff (E src, signed)	+0.24787	+0.18151	0.08534	0.06566	15048
W_eff (I src, signed)	-0.32687	-0.24028	0.08695	0.07004	14977
W_eff (E src)	+0.24787	+0.18151	0.08534	0.06566	15048
W_eff (I src)	+0.32687	+0.24028	0.08695	0.07004	14977
W_in_eff (input neurons)	+0.0013842	+0.00081433	0.1342	0.07879	6675
c_E (SFA coupling)	+0.05476	+0.082126	0.006898	0.02424	450
c_0_E (SFA offset)	-0.27431	-0.36987	0.09773	0.1679	450
a_0 (threshold)	+0.16777	+0.16904	0.08514	0.1091	300
readout_weight	-0.0057749	-0.00058804	0.1456	0.2205	6675
readout_bias	+0.0044741	+0.00010319	0.1548	0.2097	89
ic.ic (per-variant init)	-0.025816	-0.055855	0.504	0.5219	900

srnn-e-only-skip-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-e-only-skip-per-neuron)

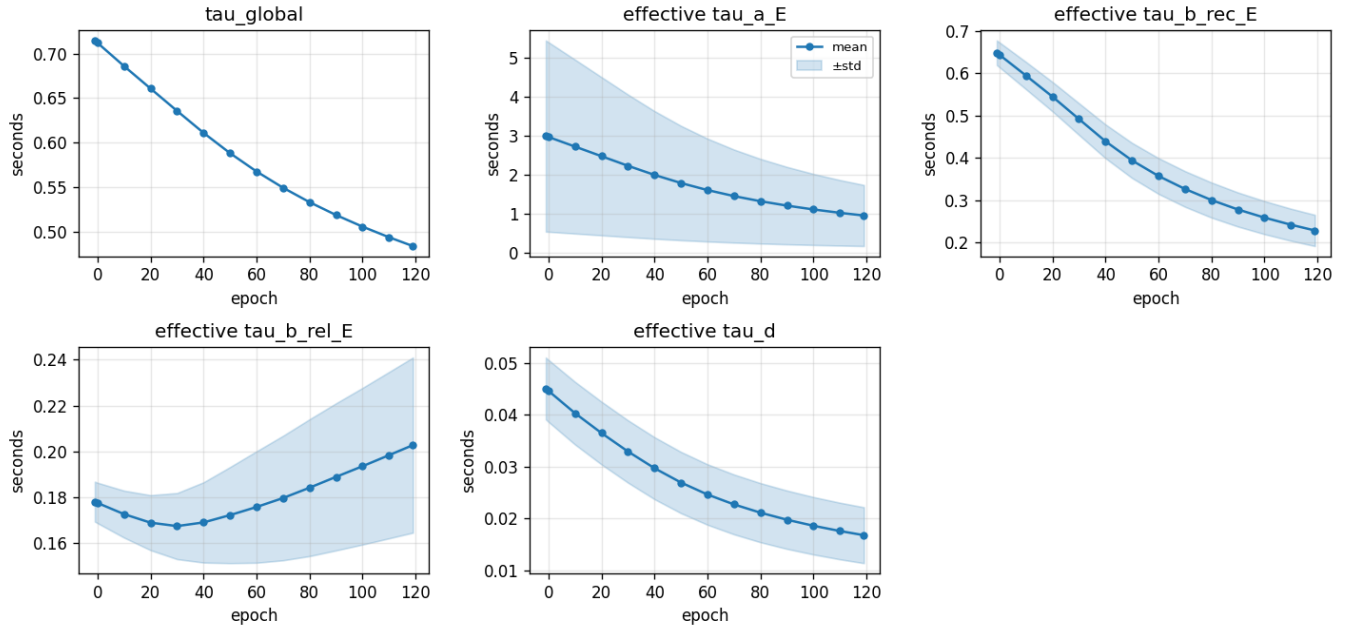
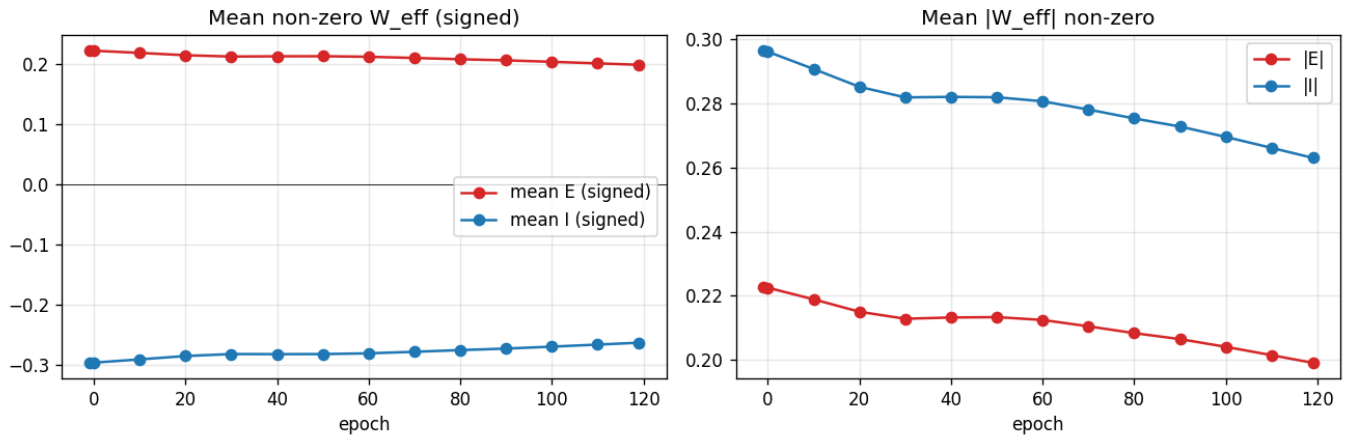


Figure 5: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

overnight-cl250-resume120 - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+0.71474	+0.48399	-	-	scalar
tau_d (s)	+0.045103	+0.016801	0.00597	0.005397	300
tau_a_E (s)	+2.9855	+0.94549	2.453	0.7839	450
tau_b_rec_E (s)	+0.64798	+0.22835	0.02907	0.0369	150
tau_b_rel_E (s)	+0.17806	+0.20273	0.008715	0.03824	150
W_eff (E src, signed)	+0.22277	+0.19911	0.07386	0.07014	15048
W_eff (I src, signed)	-0.29646	-0.26301	0.07555	0.07369	14977
W_eff (E src)	+0.22277	+0.19911	0.07386	0.07014	15048
W_eff (I src)	+0.29646	+0.26301	0.07555	0.07369	14977
W_in_eff (input neurons)	+0.001791	-0.00083735	0.0739	0.03921	6675
c_E (SFA coupling)	+0.052935	+0.099054	0.003242	0.03138	450
c_0_E (SFA offset)	-0.0012899	-0.2607	0.05942	0.1696	450
a_0 (threshold)	+0.28407	+0.14095	0.07765	0.1374	300
readout_weight	-0.013388	-0.0052303	0.07669	0.135	6675
readout_bias	+0.075552	+0.11501	0.08734	0.1561	89
ic.ic (per-variant init)	+0.062319	-0.032454	0.4358	0.5247	900

srnn-sfa-e-only-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-sfa-e-only-per-neuron)

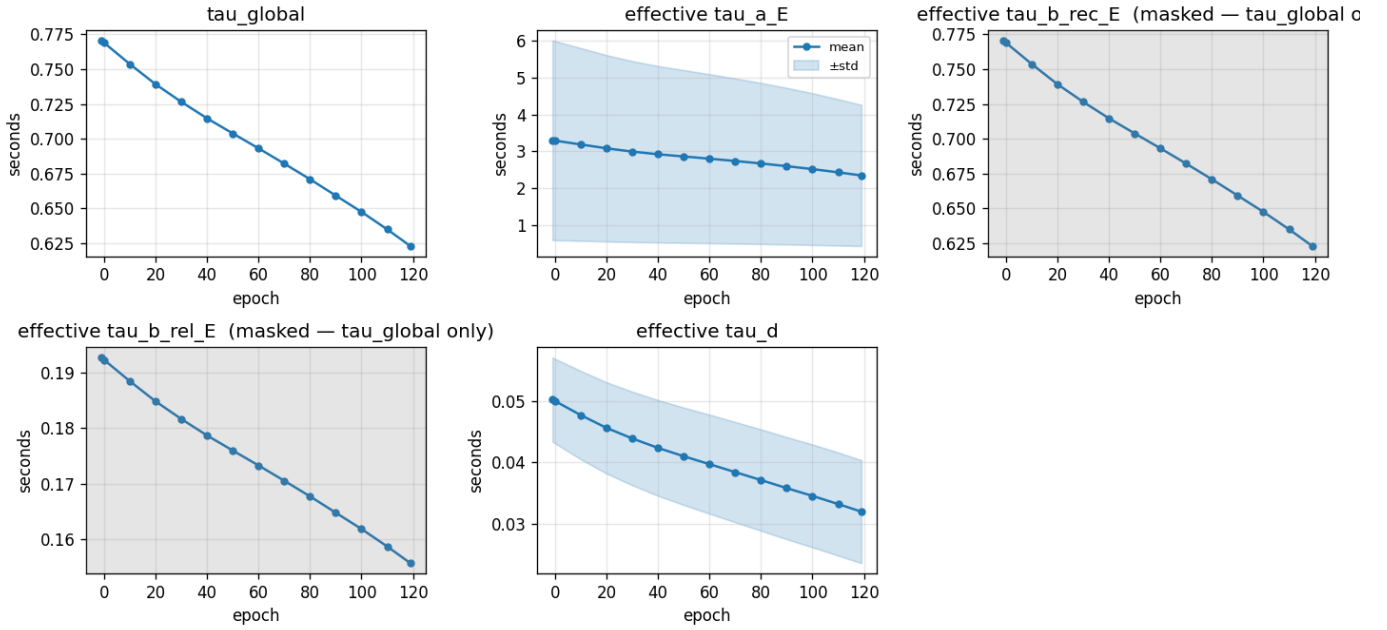
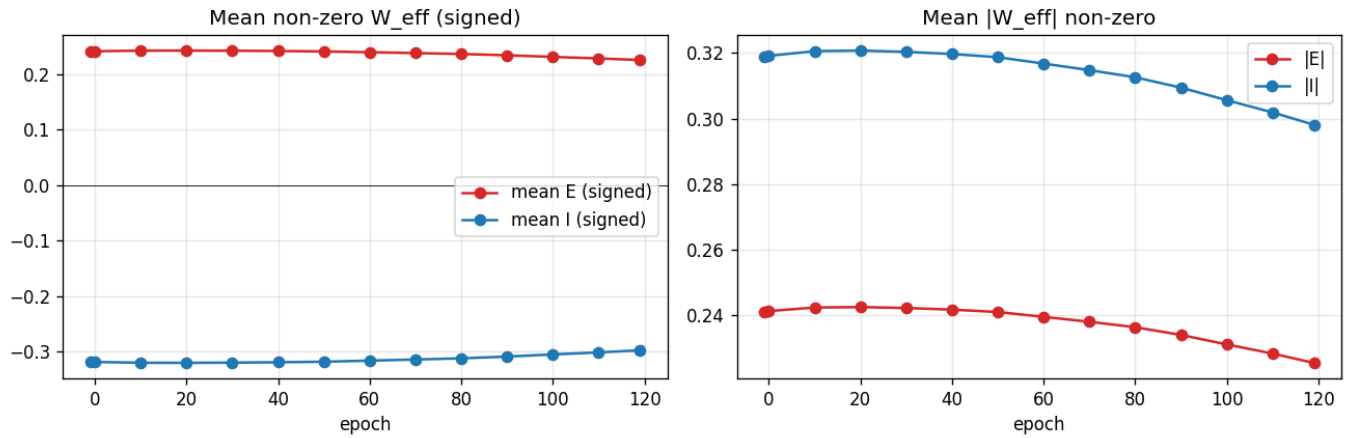


Figure 7: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# overnight-cl250-resume120 - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+0.77065	+0.6228	-	-	scalar
tau_d (s)	+0.050201	+0.031946	0.006832	0.008385	300
tau_a_E (s)	+3.2983	+2.3462	2.703	1.912	450
tau_b_rec_E (s)	+0.77065	+0.6228	0	0	150
tau_b_rel_E (s)	+0.19266	+0.1557	0	0	150
W_eff (E src, signed)	+0.24106	+0.22561	0.08131	0.07892	15048
W_eff (I src, signed)	-0.31873	-0.29814	0.08281	0.08305	14977
W_eff (E src)	+0.24106	+0.22561	0.08131	0.07892	15048
W_eff (I src)	+0.31873	+0.29814	0.08281	0.08305	14977
W_in_eff (input neurons)	-0.0020851	-0.0018244	0.1374	0.1217	6675
c_E (SFA coupling)	+0.051482	+0.045337	0.003646	0.005954	450
c_0_E (SFA offset)	-0.079732	-0.21542	0.06781	0.1249	450
a_0 (threshold)	+0.34909	+0.366	0.05572	0.08314	300
readout_weight	+0.0052418	+0.0084713	0.1206	0.1704	6675
readout_bias	-0.026226	-0.10435	0.06046	0.1289	89
ic.ic (per-variant init)	+0.063944	-0.0045598	0.4472	0.4608	900

srnn-sfa-e-only-skip-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-sfa-e-only-skip-per-neuron)

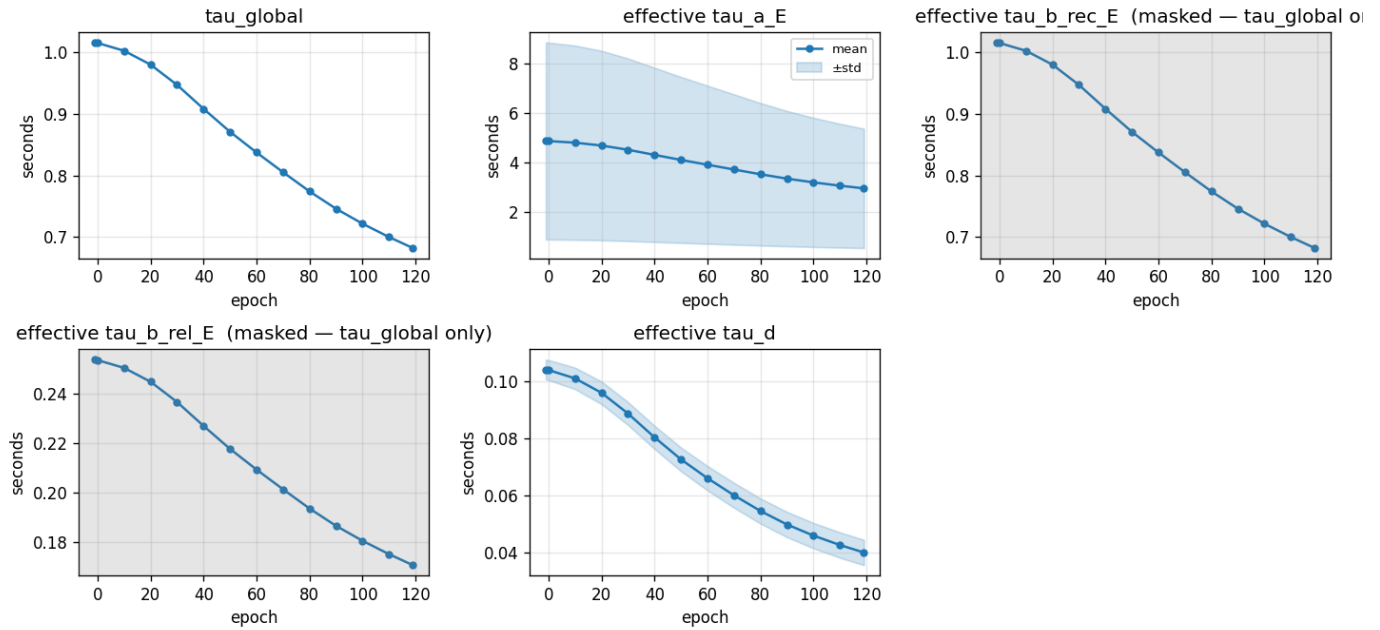
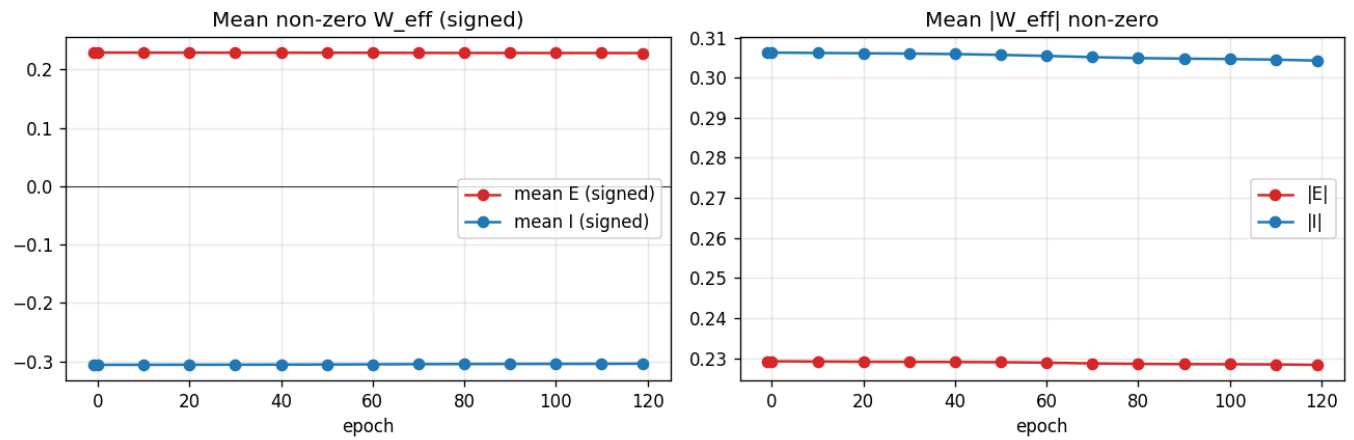


Figure 9: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# overnight-cl250-resume120 - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1.016	+0.68228	-	-	scalar
tau_d (s)	+0.10403	+0.040003	0.003474	0.004433	300
tau_a_E (s)	+4.8605	+2.9413	3.986	2.417	450
tau_b_rec_E (s)	+1.016	+0.68228	0	0	150
tau_b_rel_E (s)	+0.254	+0.17057	0	0	150
W_eff (E src, signed)	+0.22921	+0.22832	0.07535	0.07621	15048
W_eff (I src, signed)	-0.30625	-0.30425	0.07637	0.07747	14977
W_eff (E src)	+0.22921	+0.22832	0.07535	0.07621	15048
W_eff (I src)	+0.30625	+0.30425	0.07637	0.07747	14977
W_in_eff (input neurons)	-0.0014667	-0.0030177	0.09388	0.07681	6675
c_E (SFA coupling)	+0.051699	+0.053708	0.001434	0.007939	450
c_0_E (SFA offset)	+0.036936	+0.017061	0.02567	0.1012	450
a_0 (threshold)	+0.37426	+0.35718	0.03578	0.0842	300
readout_weight	-0.0010738	-0.0028867	0.06088	0.06855	6675
readout_bias	+0.0022042	+0.0371	0.003744	0.04487	89
ic.ic (per-variant init)	+0.1172	+0.054494	0.4144	0.4889	900

srnn-std-e-only-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-std-e-only-per-neuron)

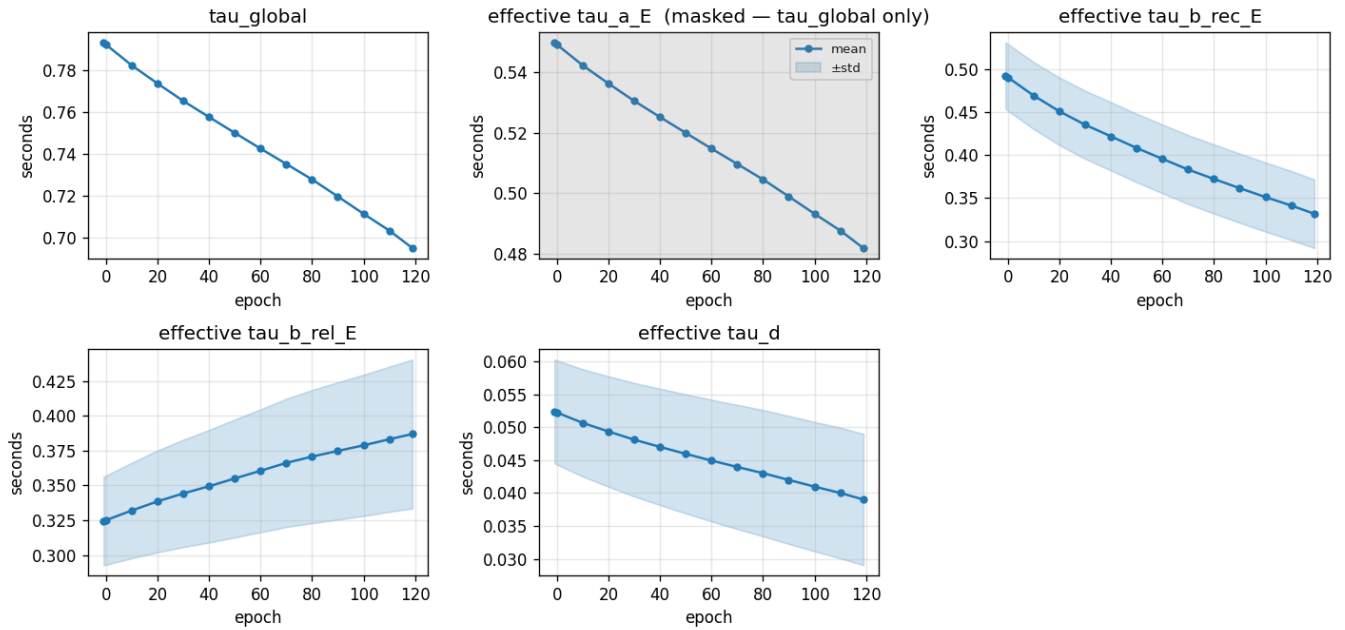


Figure 11: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-std-e-only-per-neuron)

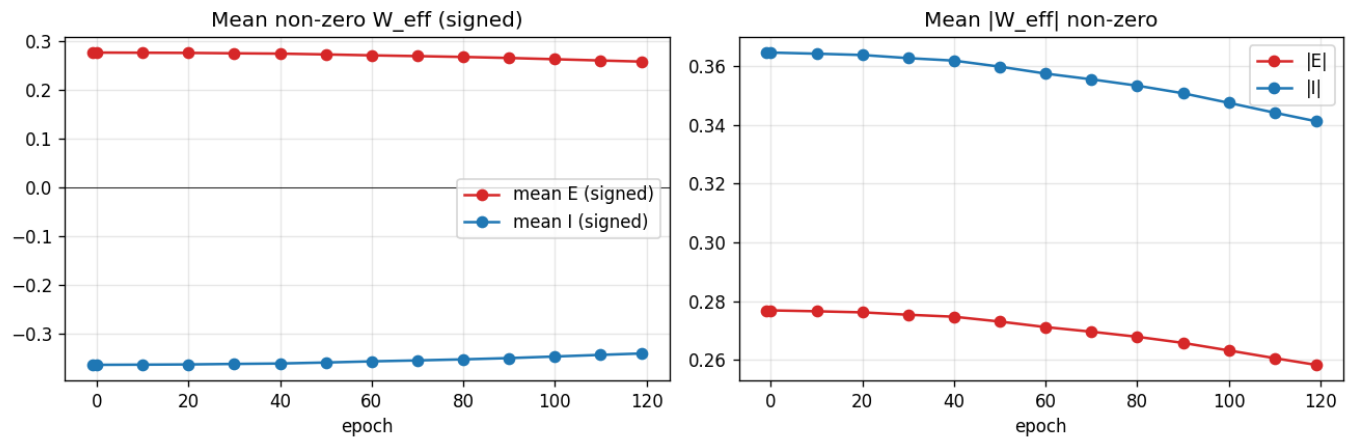


Figure 12: W_EI evolution

srnn-std-e-only-per-neuron — parameter table

overnight-cl250-resume120 - variant srnn-std-e-only-per-neuron (k=4)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+0.79317	+0.69512	-	-	scalar
tau_d (s)	+0.052401	+0.039022	0.007879	0.009972	300
tau_a_E (s)	+0.54978	+0.48182	0	0	450
tau_b_rec_E (s)	+0.49281	+0.3318	0.03874	0.03998	150
tau_b_rel_E (s)	+0.32451	+0.387	0.03178	0.05346	150
W_eff (E src, signed)	+0.27677	+0.25831	0.09482	0.09257	15048
W_eff (I src, signed)	-0.36449	-0.34123	0.09683	0.09821	14977
W_eff (E src)	+0.27677	+0.25831	0.09482	0.09257	15048
W_eff (I src)	+0.36449	+0.34123	0.09683	0.09821	14977
W_in_eff (input neurons)	-0.00091659	-0.00098562	0.1406	0.1332	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.22171	+0.28106	0.08034	0.1036	300
readout_weight	+0.0096444	+0.010267	0.1413	0.1858	6675
readout_bias	-0.022083	-0.022221	0.1547	0.187	89
ic.ic (per-variant init)	+0.0046818	+0.030821	0.5	0.4926	900

srnn-std-e-only-skip-per-neuron

overnight-cl250-resume120 — effective time constants (srnn-std-e-only-skip-per-neuron)

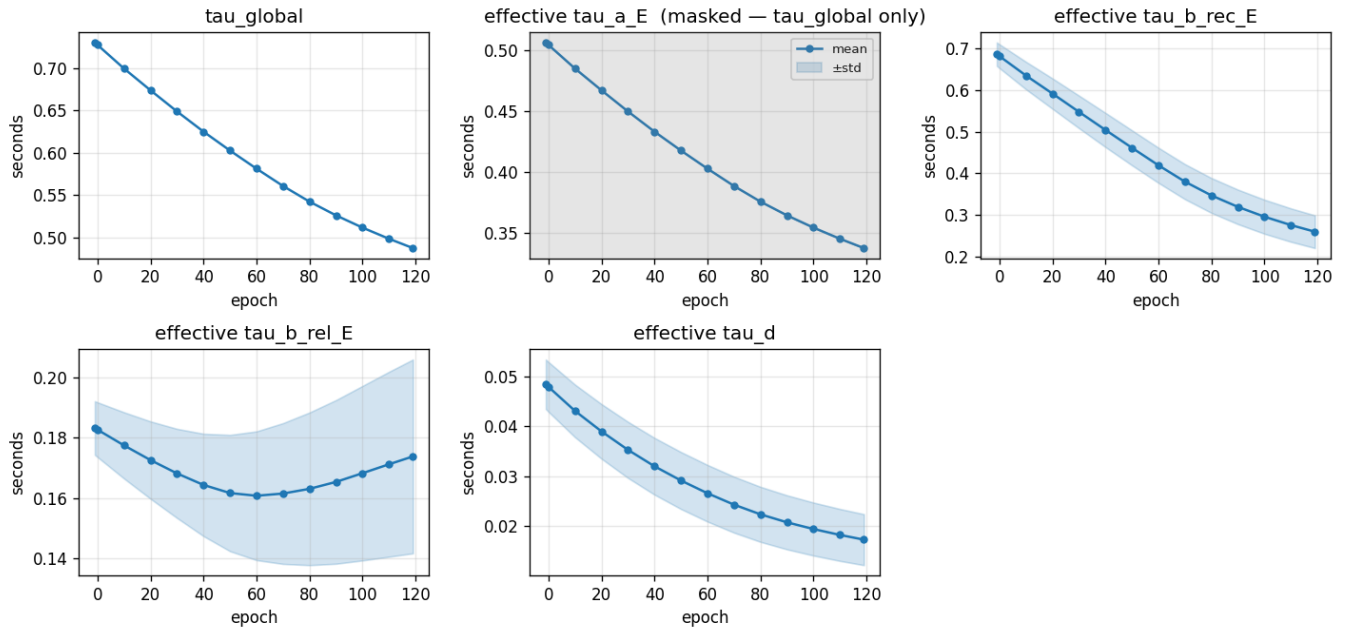


Figure 13: Tau evolution

overnight-cl250-resume120 — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

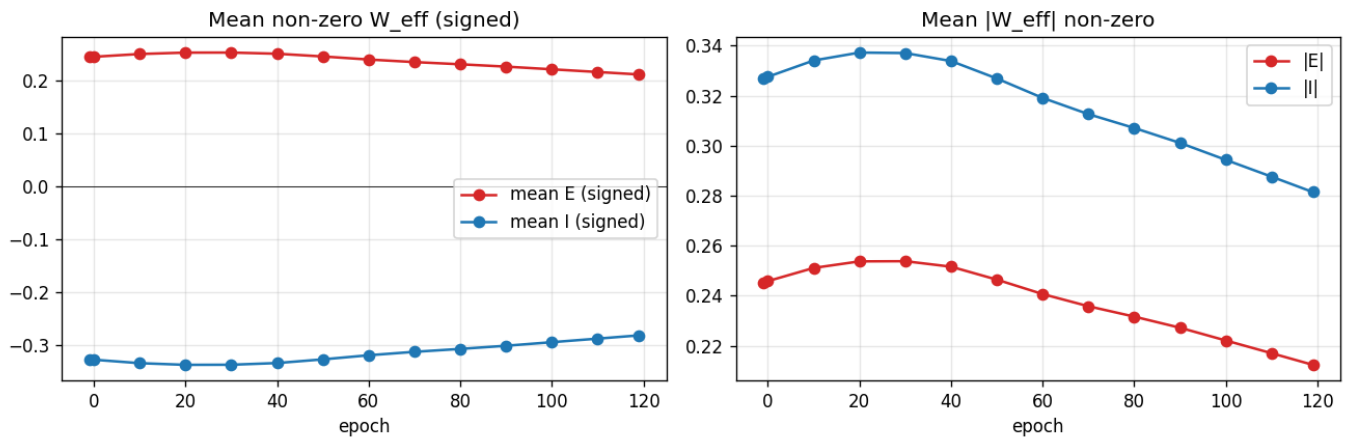


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

overnight-cl250-resume120 - variant srnn-std-e-only-skip-per-neuron (k=5)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+0.73075	+0.48709	-	-	scalar
tau_d (s)	+0.048476	+0.01728	0.004967	0.005127	300
tau_a_E (s)	+0.50652	+0.33763	0	0	450
tau_b_rec_E (s)	+0.68599	+0.25985	0.02816	0.03932	150
tau_b_rel_E (s)	+0.18328	+0.17382	0.008936	0.03227	150
W_eff (E src, signed)	+0.24529	+0.2124	0.08121	0.07436	15048
W_eff (I src, signed)	-0.3268	-0.28145	0.08245	0.07743	14977
W_eff (E src)	+0.24529	+0.2124	0.08121	0.07436	15048
W_eff (I src)	+0.3268	+0.28145	0.08245	0.07743	14977
W_in_eff (input neurons)	-0.00069182	+7.5388e-06	0.08771	0.03832	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.3437	+0.20351	0.05943	0.1406	300
readout_weight	-0.0083536	-0.0079787	0.06826	0.1213	6675
readout_bias	+0.024059	+0.050723	0.04786	0.1257	89
ic.ic (per-variant init)	+0.073535	-0.00010804	0.4271	0.4971	900